chapter (2, 3)
Anton

L-04

Techniques of Differentiation

1) $\frac{d}{dx}(uv) = u\frac{d}{dx}(v) + v\frac{d}{dx}(u)$ [product]

2) $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{d}{dx}(u) - u\frac{d}{dx}(v)}{v^2}$ [Quotient]

3) $\frac{d}{dx}(u^v) = u^v \frac{d}{dx}(v \ln u)$ [logarithmic]

4) $\frac{d}{dx}(u+v) = \frac{d}{dx}(u) + \frac{d}{dx}(v)$ [sum]

Higher Derivative

$$y = f(x)$$

$$y_1 = y' = \frac{dy}{dx} = \frac{d}{dx}[f(x)] = f'(x)$$

$$y_2 = y'' = \frac{d^2y}{dx^2} = \frac{d^2}{dx}[f(x)] = f''(x)$$

$$\vdots$$

General nth order derivative at a specific point $x = x_0$ denoted by,

$$\left.\frac{d^ny}{dx^n}\right|_{x=x_0} = f^n(x_0) = \left.\frac{d^n}{dx^n}[f(x)]\right|_{x=x_0}$$

Ex01

\# $y=(ax+b)^n$, by successive differentiation find the nth derivative.

Solution

$$y=(ax+b)^n$$

$$\frac{dy}{dx}=n(ax+b)^{n-1}\cdot a = an(ax+b)^{n-1}$$
$$= na(ax+b)^{n-1}$$

$$\frac{d^2y}{dx^2}=n(n-1)(ax+b)^{n-2}a = n(n-1)a^2(ax+b)^{n-2}$$

$$\frac{d^3y}{dx^3}=n(n-1)(n-2)a^3(ax+b)^{n-3}$$

$$\vdots$$

$$\frac{d^ny}{dx^n}=n(n-1)(n-2)\ldots(n-(n-1))a^n(ax+b)^{n-n}$$

$$=n(n-1)(n-2)\ldots 1\, a^n(ax+b)^0$$

$$=n!\,a^n\cdot 1$$

$$\boxed{\frac{d^ny}{dx^n}=n!\,a^n}$$

H.W

① $y=x^n$

② $y=\frac{1}{x+a}$

③ $y=e^{ax}$

④ $y=\sin(ax+b)$

⑤ $y=\cos(ax+b)$

must solve

→ solve sussecessive deffenentiation practice sheet.

Ex 02

~~Chain~~ Rule

$y=e^{ax}\sin(bx)$, we have to prove that

$y_2-2ay_1+(a^2+b^2)y=0$.

Solution

$y=\cancel{e}\, e^{ax}\sin bx$

$y_1=\frac{d}{dx}(e^{ax}\sin bx)$

$= ae^{ax}\sin bx + be^{ax}\cos(bx)$

$y_2=a^2e^{ax}\sin bx + bae^{ax}\cancel{\sin}\cos(bx)$

$+bae^{ax}\cos(bx) - b^2e^{ax}\sin bx$

$=(a^2-b^2)e^{ax}\sin bx + 2abe^{ax}\cos bx$

$\cancel{=(a^2}$

$\therefore y_2-2ay_1+(a^2+b^2)y$

$=(a^2-b^2)e^{ax}\sin bx + 2abe^{ax}\cos(bx)$

$-2a^2e^{ax}\sin bx - 2abe^{ax}\cos bx$

$+(a^2+b^2)e^{ax}\sin bx$

$=(a^2-b^2-2a^2+a^2+b^2)e^{ax}\sin bx$

$=0$

$\therefore y_2-2ay_1+(a^2+b^2)y=0$ (proved)

Chain rule

If $y = f(u)$, $u = g(x)$

then $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Similarly $y = f(u)$, $u = g(v)$, $v = h(x)$

then $\frac{dy}{dx} = \left(\frac{dy}{du} \cdot \frac{du}{dv} \cdot \frac{dv}{dx}\right)$

Ex 03 ~~Let,~~ $y = \ln\cos x$, find $\frac{dy}{dx}$

Let, $y = \ln u$ where $u = \cos x$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$
$$= \frac{d}{du}(\ln u) \cdot \frac{d}{dx}(\cos x)$$
$$= \frac{1}{u}(-\sin x)$$
$$= \frac{-\sin x}{\cos x} = -\tan x$$

H.W

① $y = e^{x^2}$, where, $u = x^2$, $y = e^u$

② $y = a^x$

③ $y = x^2 \cos\frac{1}{x}$

Ex 0 8 4

$y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, Find the differential coefficient of the function with respect to x. $\left(\frac{dy}{dx}\right)$

Solution

$y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, Let, ~~$x\theta = \tan\theta$~~

$x = \tan\theta$

$\Rightarrow \theta = \tan^{-1} x$

$= \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$

$= \cos^{-1}(\cos 2\theta)$

$= 2\theta$

$= 2\tan^{-1} x$

$\frac{dy}{dx} = 2\frac{d}{dx}(\tan^{-1} x) = 2\frac{1}{1+x^2} = \frac{2}{1+x^2}$

* $\tan 2A = \frac{2\tan A}{1-\tan^2 A}$

* $\sin 2A = \frac{2\tan A}{1+\tan^2 A}$

* $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$

Im

Implicit Differentiation

y is not given as a function of x explicitly.

Ex 05 Find $\frac{dy}{dx}$ at point $(1,2)$ on curve $3x^4 - x^2y + 2y^3 = 0$.

solution

$$3x^4 - x^2y + 2y^3 = 0$$

$\therefore$ differentiate with respect to x,

$$\frac{d}{dx}[3x^4 - x^2y + 2y^3] = 0$$

$$\Rightarrow \frac{d}{dx}[3x^4] - \frac{d}{dx}[x^2y] + \frac{d}{dx}[2y^3] = 0$$

$$\Rightarrow 12x^3 - x^2\frac{dy}{dx} - 2xy + 6y^2\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx}(6y^2 - x^2) = 2xy - 12x^3$$

$$\therefore \frac{dy}{dx} = \frac{2xy - 12x^3}{6y^2 - x^2}$$

at point $(1,2)$

$$\left.\frac{dy}{dx}\right|_{(1,2)} = \frac{2\times 1\times (2) - 12(1)^3}{6(2)^2 - (1)^2}$$

$$= \frac{4-12}{24-1} = \frac{-8}{23} \quad \text{(Ans)}$$

Logarithmic differentiation

Ex 06

$x^y = y^x$, find $\frac{dy}{dx}$.

Solution

$$x^y = y^x$$

$$\Rightarrow \ln(x^y) = \ln(y^x)$$

$$\Rightarrow y \ln x = x \ln y$$

$\therefore$ differentiate,

$$\frac{d}{dx}(y \ln x) = \frac{d}{dx}(x \ln y)$$

$$\Rightarrow \frac{dy}{dx} \ln x + y\frac{1}{x} = \ln y + x\frac{1}{y}\frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx}\left(\ln x - \frac{x}{y}\right) = \ln y - y/x$$

$$\Rightarrow \frac{dy}{dx} = \frac{\ln y - y/x}{\ln x - x/y}$$ (Ans.)

H.W

1) $\cos(x+y) = e^{x+y}$,

2) $xy + e^y = 0$, } find $\frac{dy}{dx}$

3) $(\sin x)^{\cos x} + (\cos x)^{\sin x}$ - ~~using implicite~~

rule

* $\ln(xy) = \ln x + \ln y$
* $\ln\left(\frac{x}{y}\right) = \ln x - \ln y$

Ex07

$y = x^{\sin x}$; Find $\frac{dy}{dx}$

$\sinh x = \frac{e^x - e^{-x}}{2}$

$\cosh x = \frac{e^x + e^{-x}}{2}$

$\frac{d}{dx}(\cos x) = -\sin x$

$\frac{d}{dx}(\cosh x) = \sinh x$

Solution:

$y = x^{\sin x}$

$\Rightarrow \ln y = \cancel{\ln(x^{\sin x})}\ \ln(x^{\sin x})$

$\Rightarrow \ln y = \sin x \ln x$

Differentiate,

$\frac{d}{dx}(\ln y) = \frac{d}{dx}(\sin x \ln x)$

$\Rightarrow \frac{1}{y} \frac{dy}{dx} = \sin x \frac{1}{x} + \cos x \ln x$

$\Rightarrow \frac{dy}{dx} = y\left(\frac{\sin x}{x} + \ln x \cos x\right)$

$\therefore \frac{dy}{dx} = x^{\sin x}\left(\frac{\sin x}{x} + \ln x \cos x\right)$

Ex08 $y = x^2 e^x \sin x \cosh x$. Find $\frac{dy}{dx}$

$\ln y = \ln(x^2 e^x \sin x \cosh x)$ $\boxed{\ln e = 1}$

$\Rightarrow \ln y = \ln x^2 + \ln(e^x) + \ln(\sin x) + \ln(\cosh x)$

$\Rightarrow \ln y = 2\ln x + x\ln e + \ln(\sin x) + \ln(\cosh x)$

$\therefore \frac{1}{y}\frac{dy}{dx} = \frac{2}{x} + 1 + \frac{\cos x}{\sin x} + \frac{1}{\cosh x} \sinh x$

$\Rightarrow \frac{dy}{dx} = y\left[\frac{2}{x} + 1 + \cot x + \tanh x\right]$

$= x^2 e^x \sin x \cosh x \left[\frac{2}{x} + 1 + \cot x + \tanh x\right]$

CamScanner

* Parametric Differentiation:

Ex 08 $x = a\cos\theta$, $y = b\sin\theta$; $0 \le \theta \le 2\pi$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \Big/ \frac{dx}{d\theta} = \frac{b\cos\theta}{-a\sin\theta} = \frac{-b}{a}\cot\theta$$

Ex 09 $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$

Find $\frac{dy}{dx}$.

Solution

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

$$= \frac{a(\sin\theta)}{a(1+\cos\theta)} = \frac{\sin\theta}{1+\cos\theta}$$

Ex 10. $x = \sin^2\theta$, $y = \tan\theta$, $\frac{dy}{dx} = ?$

$= (\sin\theta)^2$

Solution

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{\sec^2\theta}{2\sin\theta\cos\theta}$$

(Ans)

Differentiate $f(x)$ with respect to $g(x)$

$y = f(x)$, $z = g(x)$

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}}$$

Ex 11.1 Differentiate $x^{\sin^{-1}x}$ with respect to $\sin^{-1}x$.

Solution: Let, $y = x^{\sin^{-1}x}$, $z = \sin^{-1}x$

$$\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{\frac{dy}{dx}}{\frac{dz}{dx}}$$

$$\frac{dz}{dx} = \frac{d}{dx}(\sin^{-1}x) = \frac{1}{\sqrt{1-x^2}}$$

Here, $y = x^{\sin^{-1}x}$

$\Rightarrow \ln y = \ln (x^{\sin^{-1}x})$

$\Rightarrow \ln y = \sin^{-1}x \cdot \ln(x)$

Differentiate, $\frac{1}{y}\frac{dy}{dx} = \frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}$

$\Rightarrow \frac{dy}{dx} = y\left[\frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right]$

$\therefore \frac{dy}{dx} = x^{\sin^{-1}x}\left[\frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right]$

$$\therefore \frac{dy}{dz} = \frac{x^{\sin^{-1}x}\left[\frac{\sin^{-1}x}{x} + \frac{\ln x}{\sqrt{1-x^2}}\right]}{\frac{1}{\sqrt{1-x^2}}}$$

(Ans)

$[\ln(xy) = \ln x + \ln y]$

$[\ln(\frac{x}{y}) = \ln x - \ln y]$

Ex 12

Q. $\ln\sqrt{\frac{1+\sin x}{1-\sin x}}$; find the differential coefficient.

Solution

Let, $y = \ln\sqrt{\frac{1+\sin x}{1-\sin x}}$

$\Rightarrow y = \ln\left(\frac{1+\sin x}{1-\sin x}\right)^{1/2}$

$\Rightarrow y = \frac{1}{2}\ln\left(\frac{1+\sin x}{1-\sin x}\right)$

$\Rightarrow \frac{dy}{dx} = \frac{d}{dx}\left[\frac{1}{2}\ln\left(\frac{1+\sin x}{1-\sin x}\right)\right]$

$= \frac{1}{2}\left[\frac{d}{dx}[\ln(1+\sin x) - \ln(1-\sin x)]\right]$

$= \frac{1}{2}\left[\frac{1}{1+\sin x}(\cos x) - \frac{1}{1-\sin x}(-\cos x)\right]$

$= \frac{1}{2}\left[\frac{\cos x}{1+\sin x} + \frac{\cos x}{1-\sin x}\right]$

$= \frac{1}{2}\left[\frac{\cos x(1-\sin x) + \cos x(1+\sin x)}{(1+\sin x)(1-\sin x)}\right]$

$= \frac{1}{2}\left[\frac{\cos x - \cos x\sin x + \cos x + \cos x\sin x}{1-\sin^2 x}\right]$

$= \frac{1}{2} \times \frac{2\cos x}{\cos^2 x}$

$= \frac{1}{\cos x} = \sec x$ (Ans)

$[\cancel{\ln(x+y)}]$
$= [\cancel{\ln x + \ln y}]$

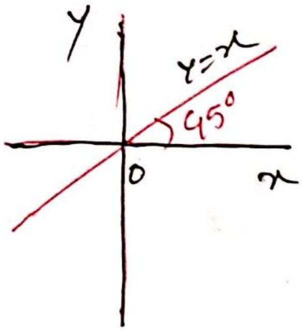
$[\ln(\frac{x}{y})]$
$= [\ln x - \ln y]$

$[\sin^2 x + \cos^2 x = 1]$
$\Rightarrow [\cos^2 x = 1 - \sin^2 x]$

Sketch

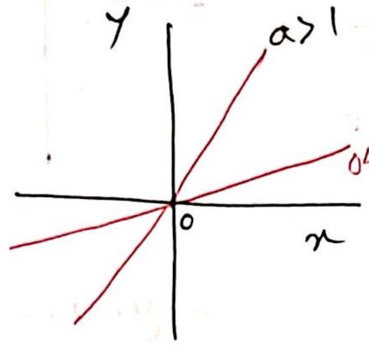
$f(x) = x$

$y = x$



* $f(x) = ax$

$y = ax$



* $f(x) = mx + c$

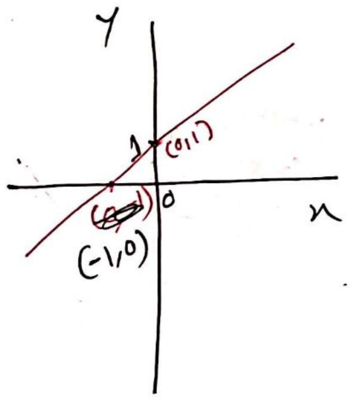
$f(x) = x + 1$

$y = x + 1$

$\Rightarrow (y - 1) = x$

$\left[\begin{array}{l} y = 0, \\ x = -1 \end{array} \right.$

$\left[\begin{array}{l} x = 0, \\ y = 1 \end{array} \right.$



* $f(x) = x - 1$

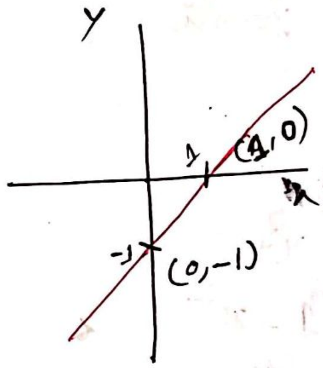
$y = x - 1$

$\Rightarrow y + 1 = x$

if

$\left[\begin{array}{l} y = 0 \\ x = 1 \end{array} \right.$

$\left[\begin{array}{l} x = 0 \\ y = -1 \end{array} \right.$



* $y - 2 = x + 3$

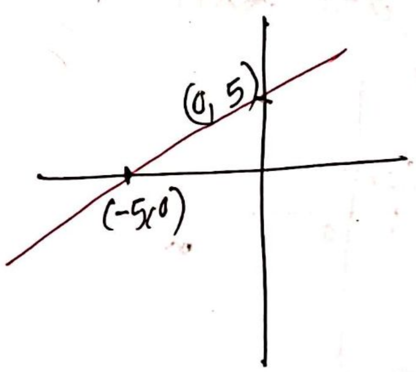
$\Rightarrow y - 2 - 3 = x$

$\Rightarrow y - 5 = x$

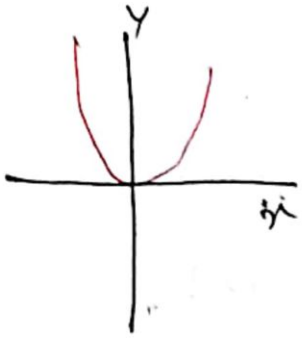
if

$\left[\begin{array}{l} x = 0, \\ y = 5 \end{array} \right.$

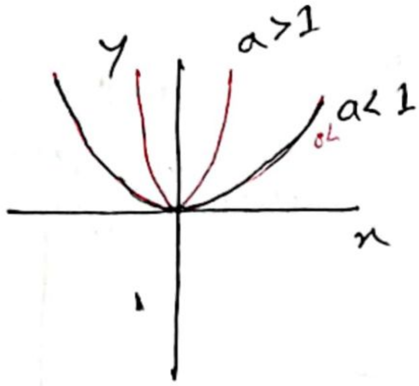
$\left[\begin{array}{l} \text{if } y = 0 \\ x = -5 \end{array} \right.$



* $y = x^2$

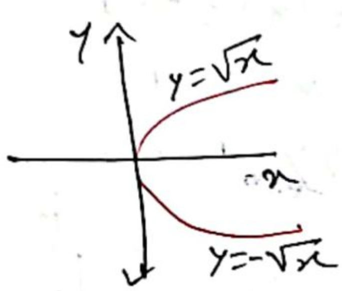


* $y = ax^2$



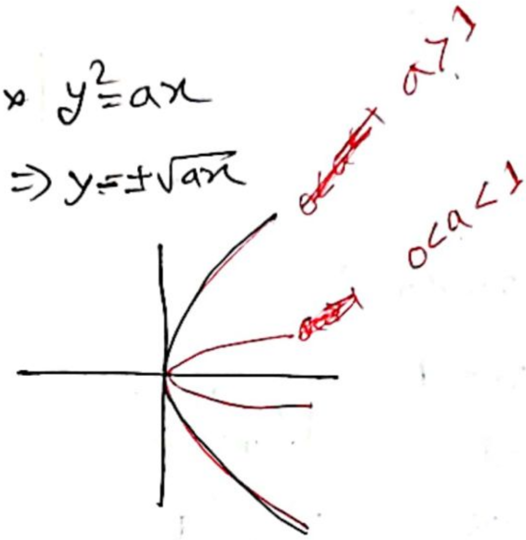
* $y^2 = x$

$\Rightarrow y = \pm\sqrt{x}$



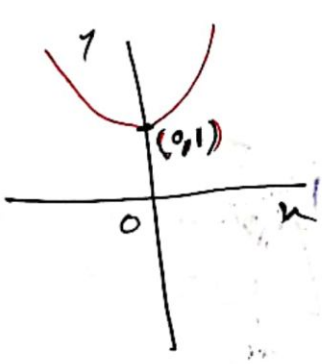
* $y^2 = ax$

$\Rightarrow y = \pm\sqrt{ax}$



* $y = x^2 + 1$

$\Rightarrow y - 1 = x^2$



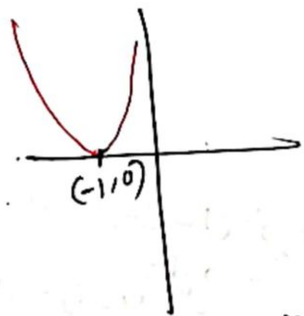
$y = (x+1)^2$

$y = 0,\ (x+1)^2 = 0$

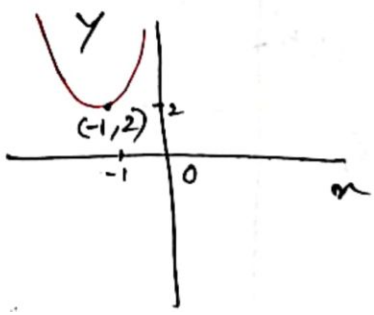
$(x+1) = 0$

$\Rightarrow x+1 = 0$

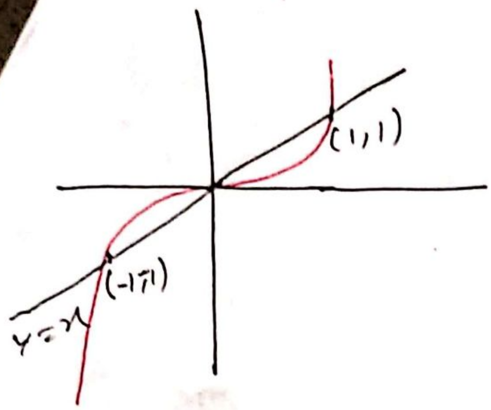
$x = -1$



* $y - 2 = (x+1)^2$

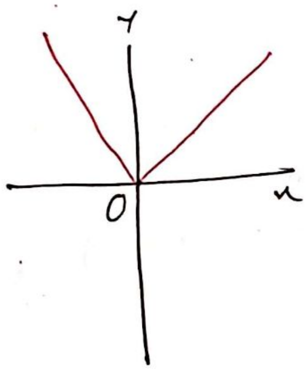


* $y = x^3$

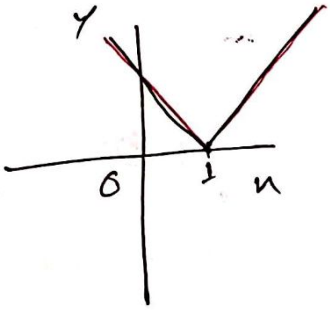


*

* $y = |x|$

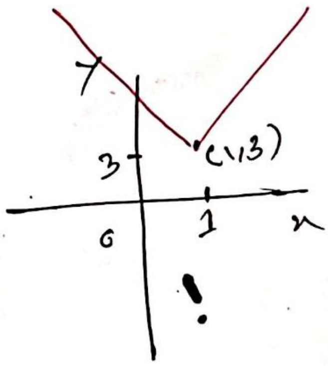


* $y = |x-1|$



* $y = |x-1| + 3$

$\Rightarrow y - 3 = |x-1|$



* $y = |x+1| - 3$

$\Rightarrow y + 3 = |x+1|$